Title

Pump P-101 Centrifugal Pump Maintenance Manual

Document Information

Field	Value
Document Number	PM-P101-001
Equipment Tag	P-101
Equipment Name	Centrifugal Process Pump
Area	Crude Oil Transfer Station
Department	Maintenance
Revision	Rev-4
Issue Date	15 March 2024
Approved By	Maintenance Manager
Standard	API 610
Quality Standard	ISO 9001
Safety Standard	OISD-STD-118

---

1. Purpose

This manual describes the preventive maintenance, inspection procedures, troubleshooting guidelines and safe operation of Pump P-101 installed in the Crude Oil Transfer Station.

The objective is to maximize equipment reliability while minimizing unplanned shutdowns.

---

2. Equipment Description

Pump Tag:
P-101

Motor Tag:
M-101

Isolation Valve:
V-201

Pump Type:
Horizontal Centrifugal Pump

Capacity:
250 m3/hr

Operating Pressure:
12 Bar

Operating Temperature:
90 °C

Fluid:
Crude Oil

---

3. Safety Precautions

WARNING

Always isolate electrical supply before maintenance.

Apply Lock Out Tag Out (LOTO).

Depressurize pipeline before opening flanges.

Wear PPE.

Use flame resistant clothing.

Use safety goggles.

Use gloves.

Never operate pump without lubrication.

Emergency shutdown shall be followed during abnormal vibration.

---

4. Preventive Maintenance SOP

Daily

• Check vibration

• Check leakage

• Check oil level

Weekly

• Inspect coupling

• Inspect bearing temperature

Monthly

• Lubricate bearings

• Inspect seals

Quarterly

• Alignment inspection

• Replace damaged gasket

Half Yearly

• Complete inspection

• Shaft runout measurement

Annually

• Overhaul

• Replace bearings

• Replace mechanical seal

---

5. Inspection Checklist

□ Leakage

□ Pressure

□ Temperature

□ Bearing Noise

□ Seal Condition

□ Coupling Alignment

□ Lubrication

□ Motor Current

□ Foundation Bolts

---

6. Failure Modes

High vibration

Bearing failure

Seal leakage

Impeller erosion

Misalignment

Motor overload

Low suction pressure

Excessive temperature

---

7. Root Cause Analysis

Bearing Failure

Possible Causes

Poor lubrication

Misalignment

Water contamination

Recommended Action

Replace bearing

Flush lubrication system

Check alignment

Seal Leakage

Possible Causes

Mechanical seal damage

Pressure surge

Improper installation

Recommended Action

Replace seal

Inspect shaft sleeve

Verify operating pressure

---

8. Spare Parts

Bearing SKF-6312

Mechanical Seal Type-A

Coupling Element CE-12

Oil Seal OS-17

Impeller IP-101

Gasket Set GS-101

---

9. Compliance

This maintenance procedure complies with:

API 610

ISO 9001

OISD-STD-118

Factory Safety Requirements

10. Lessons Learned

Previous incidents identified inadequate lubrication as the primary contributor to bearing failure.

Maintenance interval was reduced from 1000 hours to 500 hours.

Regular vibration monitoring reduced unexpected shutdowns by approximately 28%.

11. Revision History

Rev-1 Initial Release

Rev-2 Inspection Checklist Updated

Rev-3 Added Seal Maintenance

Rev-4 Updated Preventive Maintenance Schedule